Problem Set 4

AI 501: Mathematics for Artificial Intelligence | Fall 2026

Problem 1

Let

$$A = \begin{pmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{pmatrix}.$$

(a) Reduce A to echelon form. Find $\text{rank}(A)$, the pivot and free variables, and a basis for $\mathcal{N}(A)$.

(b) Find the condition on $b = (b_1, b_2, b_3)^T$ for $Ax = b$ to be solvable.

(c) For $b = (1, 5, 6)^T$, find the complete solution $x = x_p + x_n$.

(d) The lecture slide says "if $\text{rank}(A) < n$, the system has infinitely many solutions." Use this example to state the claim correctly.

Solution:

(a) $R_2 - 2R_1 \to (0, 0, 2, 4)$, $R_3 - 3R_1 \to (0, 0, 2, 4)$, then $R_3 - R_2 \to 0$:

$$U = \begin{pmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Pivots are in columns 1 and 3, so $\text{rank}(A) = 2$; x_2, x_4 are free. Setting $(x_2, x_4) = (1, 0)$ gives $s_1 = (-2, 1, 0, 0)^T$; setting $(x_2, x_4) = (0, 1)$ gives $s_2 = (2, 0, -2, 1)^T$. Hence $\mathcal{N}(A) = \text{span}\{s_1, s_2\}$ and $\dim \mathcal{N}(A) = 4 - 2 = 2$.

(b) The same elimination on $[A \mid b]$ leaves the last row $[\,0\ 0\ 0\ 0 \mid b_3 - b_2 - b_1\,]$. So $Ax = b$ is solvable iff $b_3 = b_1 + b_2$; geometrically, $\mathcal{C}(A)$ is the plane $b_1 + b_2 - b_3 = 0$ in $\mathbb{R}^3$.

(c) Since $1 + 5 = 6$, the system is solvable. The reduced equations are $x_1 + 2x_2 + 2x_3 + 2x_4 = 1$ and $2x_3 + 4x_4 = 3$. Setting the free variables to 0 gives $x_3 = 3/2$ and $x_1 = -2$, so

$$x = \begin{pmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{pmatrix} + c_1 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 0 \\ -2 \\ 1 \end{pmatrix}, \qquad c_1, c_2 \in \mathbb{R}.$$

(d) $r < n$ means $\mathcal{N}(A)$ is nontrivial, which gives infinitely many solutions *only if* $b \in \mathcal{C}(A)$. For $b = (1, 5, 7)^T$ there is no solution. The correct statement: if $r < n$, there are either 0 or infinitely many solutions.

Problem 2

Let A be $m \times n$ with rank r. For each case, state how many solutions $Ax = b$ can have, and give a small example matrix.

(i) $r = m = n$

(ii) $r = n < m$

(iii) $r = m < n$

(iv) $r < m$ and $r < n$

Solution:

(i) Exactly one solution for every b (A is square and invertible). Example: $\begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$.

(ii) 0 or 1 solutions: $\mathcal{N}(A) = \{0\}$, but $\mathcal{C}(A) \neq \mathbb{R}^m$. Example: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$.

(iii) Infinitely many solutions for every b: $\mathcal{C}(A) = \mathbb{R}^m$ and $\mathcal{N}(A)$ is nontrivial. Example: $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

(iv) 0 or infinitely many solutions. Example: the matrix in Problem 1.

Problem 3

Construct a matrix with each property, or explain why it is impossible.

(a) Its column space contains $(1, 1, 0)^T$ and $(0, 0, 1)^T$, and its row space contains $(1, 2)^T$ and $(2, 5)^T$.

(b) Its column space has basis $\{(1, 1, 3)^T\}$, and its null space has basis $\{(3, 1, 1)^T\}$.

(c) $\dim \mathcal{N}(A) = 1 + \dim \mathcal{N}(A^T)$.

Solution:

(a) **Possible.** The matrix must be 3×2 with rank 2; then the row space is all of $\mathbb{R}^2$ and contains both vectors. Example: $A = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(b) **Impossible.** The column basis gives $m = 3$ and $r = 1$. The null-space vector lies in $\mathbb{R}^3$, so $n = 3$ and nullity $= 1$. Rank–nullity would require $1 + 1 = 3$.

(c) **Possible.** We need $n - r = 1 + (m - r)$, i.e. $n = m + 1$. Example: $A = \begin{pmatrix} 1 & 0 \end{pmatrix}$, with $\dim \mathcal{N}(A) = 1$ and $\dim \mathcal{N}(A^T) = 0$.

Problem 4

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$.

(a) Show that $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$ and $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$.

(b) Hence prove $\operatorname{rank}(AB) \leq \min(\operatorname{rank} A, \operatorname{rank} B)$.

(c) Give 2×2 matrices for which the inequality is strict.

Solution:

(a) *Column space.* Let $a_1, \ldots, a_n$ be the columns of A and b_j the j-th column of B. The j-th column of AB is

$$Ab_j = b_{1j}a_1 + b_{2j}a_2 + \cdots + b_{nj}a_n,$$

a linear combination of the columns of A. So every column of AB lies in $\mathcal{C}(A)$, and hence so does every combination of them: $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$.

Concrete check: $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ gives $AB = \begin{pmatrix} 3 & 0 \\ 7 & 0 \end{pmatrix}$, and $(3, 7)^T = 1 \cdot (1, 3)^T + 1 \cdot (2, 4)^T$. Multiplying by B only mixes the columns of A; it cannot produce a direction outside $\mathcal{C}(A)$.

Null space. If $Bx = 0$, then $ABx = A0 = 0$, so $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$.

(b) From (a), $\operatorname{rank}(AB) = \dim \mathcal{C}(AB) \leq \dim \mathcal{C}(A) = \operatorname{rank} A$. Also $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$ gives $\operatorname{nullity}(AB) \geq \operatorname{nullity}(B)$. Since AB and B both have p columns, rank–nullity gives $\operatorname{rank}(AB) = p - \operatorname{nullity}(AB) \leq p - \operatorname{nullity}(B) = \operatorname{rank} B$.

(c) $A = B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$: each has rank 1, but $AB = 0$ has rank 0.

Problem 5

Suppose every label is predicted by a single constant c, given data $y_1, \ldots, y_n$.

(a) Show that the c minimizing $\frac{1}{n}\sum_i (y_i - c)^2$ is the sample mean.

(b) Show that the c minimizing $\frac{1}{n}\sum_i |y_i - c|$ is a sample median.

(c) For $y = \{2, 3, 4, 5, 100\}$, find both minimizers. Use the result to explain the lecture's remark that MAE is suited for noisy data.

Solution:

(a) $\frac{d}{dc}\frac{1}{n}\sum_i (y_i - c)^2 = -\frac{2}{n}\sum_i (y_i - c) = 0 \Rightarrow c = \bar{y}$. The second derivative is $2 > 0$, so this is a minimum.

(b) The loss is not differentiable at the data points, so use its slope. For c between data points,

$$\frac{d}{dc}\sum_i |y_i - c| = \#\{y_i < c\} - \#\{y_i > c\}.$$

The loss decreases while more points lie above c and increases once more lie below. The minimum is where half the points are on each side of c: the median.

(c) Mean = 22.8, median = 4. The single outlier drags the squared-loss solution far from the other four points, while the absolute-loss solution barely moves. Squared loss grows quadratically, so outliers dominate it; absolute loss grows linearly, making it more robust to noise.

Problem 6

The training data are:

Obs	X_1	X_2	X_3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	−1	0	1	Green
6	1	1	1	Red

We want to predict Y at the test point $X_1 = X_2 = X_3 = 0$.

(a) Compute the Euclidean distance from each observation to the test point.

(b) What is the prediction with $k = 1$? Why?

(c) What is the prediction with $k = 3$? Why?

(d) If the class decision boundary in this problem is highly nonlinear, would you expect the best k to be large or small? Why?

Solution:

(a) Obs 1: 3; Obs 2: 2; Obs 3: $\sqrt{10} \approx 3.16$; Obs 4: $\sqrt{5} \approx 2.24$; Obs 5: $\sqrt{2} \approx 1.41$; Obs 6: $\sqrt{3} \approx 1.73$.

(b) **Green.** The nearest point is Obs 5, which is Green.

(c) **Red.** The three nearest points are Obs 5 (Green), Obs 6 (Red), and Obs 2 (Red); the majority is Red.

(d) **Small.** A small k gives a flexible, high-variance boundary that can follow nonlinear shapes. A large k averages over large neighborhoods, smoothing the boundary toward a nearly linear shape.

Problem 7

Consider brute-force kNN with n training points in $\mathbb{R}^d$.

(a) Give the time complexity of classifying one test point, broken down into computing distances, selecting the k nearest, and voting. Also give the storage cost. For MNIST ($n = 60{,}000$, $d = 784$), roughly how many multiply-adds does one query need?

(b) Why is an odd k recommended for binary classification? Does an odd k prevent ties when there are three classes?

(c) A dataset records each person's annual salary in dollars and age in years. When kNN uses Euclidean distance, a difference of \$1,000 in salary counts far more than a difference of 50 years in age. Explain why, and give the fix.

Solution:

(a) Computing distances: $O(nd)$. Selecting the k nearest: $O(n)$ with quickselect, $O(n \log k)$ with a heap, or $O(n \log n)$ with a full sort. Voting: $O(k)$. The total is $O(nd)$ per query, and $O(mnd)$ for m test points. Storage is $O(nd)$, because the whole training set must be kept. There is no training cost; all the cost moves to prediction time. For MNIST, one query needs about $60{,}000 \times 784 \approx 4.7 \times 10^7$ multiply-adds.

(b) With two classes and odd k, one class always has a strict majority. With three classes and $k = 3$, the neighbors can be one of each class, so ties are still possible. The fallbacks from lecture apply: use $k - 1$ neighbors, fall back to 1-NN, or use class priors.

(c) Euclidean distance adds squared differences across features, so a feature measured in large units (dollars) dominates one measured in small units (years), regardless of which is more informative. The fix is to normalize: standardize each feature to mean 0 and variance 1, or rescale it to $[0, 1]$, using statistics computed on the training set only.

Problem 8

A screening test on 2030 people gives $TP = 20$, $FP = 180$, $FN = 10$, $TN = 1820$.

(a) Compute sensitivity, specificity, FPR, precision, negative predictive value NPV $= \frac{TN}{TN+FN}$, and accuracy.

(b) The test has 91% specificity, yet only 10% of positive results are true positives. Explain why.

Solution:

(a) Sensitivity $= 20/30 \approx 0.667$; Specificity $= 1820/2000 = 0.91$; FPR $= 180/2000 = 0.09$;
Precision $= 20/200 = 0.10$; NPV $= 1820/1830 \approx 0.995$;
Accuracy $= 1840/2030 \approx 0.906$.

(b) Only 30 of 2030 people have the condition (prevalence $\approx 1.5\%$). A 9% FPR applied to 2000 healthy people produces 180 false positives, which swamp the 20 true positives. Precision depends on prevalence; sensitivity and specificity do not. The test is still useful for screening because its NPV is very high: a negative result is almost always correct.

Problem 9

The Matthews correlation coefficient is

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}.$$

(a) A dataset has 91 positives and 9 negatives. A classifier gives $TP = 90$, $FN = 1$, $TN = 0$, $FP = 9$. Compute accuracy, F_1, and MCC. Which metric tells the truth about this classifier?

(b) Compute F_1 and MCC for the lecture's heart-disease example: $TP = 100$, $FP = 10$, $FN = 5$, $TN = 50$.

Solution:

(a) Accuracy $= 90/100 = 0.90$. $F_1 = \dfrac{180}{190} \approx 0.95$. $\text{MCC} = \dfrac{0 - 9}{\sqrt{99 \cdot 91 \cdot 9 \cdot 1}} = \dfrac{-9}{284.7} \approx -0.03$.

The classifier never identifies a single negative. MCC correctly reports that it is no better than chance. F_1 ignores TN, and accuracy is inflated by the class imbalance.

(b) $F_1 = \dfrac{200}{215} \approx 0.93$. $\text{MCC} = \dfrac{5000 - 50}{\sqrt{110 \cdot 105 \cdot 60 \cdot 55}} = \dfrac{4950}{6173.7} \approx 0.80$.

MCC is high only when all four cells of the confusion matrix are good. It is slightly lower than F_1 here because it also accounts for the FP/TN performance.

Problem 10

Twenty test instances sorted by classifier score (p = actually positive, n = actually negative):

0.9	0.8	0.7	0.6	0.55	0.54	0.53	0.52	0.51	0.505
p	p	n	p	p	p	n	n	p	n
0.4	0.39	0.38	0.37	0.36	0.35	0.34	0.33	0.30	0.1
p	n	p	n	n	n	p	n	p	n

(a) An instance is predicted positive if its score is at least t. Give the (FPR, TPR) point for $t = 0.54$, 0.51, and 0.38.

(b) Compute the accuracy at $t = 0.54$.

(c) Compute the AUC, using the interpretation that AUC is the probability that a randomly chosen positive is scored higher than a randomly chosen negative.

(d) Where do the perfect classifier and a random classifier lie in ROC space?

Solution: There are 10 positives and 10 negatives.

(a) $t = 0.54$: $TP = 5$, $FP = 1 \Rightarrow (0.1, 0.5)$. $t = 0.51$: $TP = 6$, $FP = 3 \Rightarrow (0.3, 0.6)$. $t = 0.38$: $TP = 8$, $FP = 5 \Rightarrow (0.5, 0.8)$.

(b) At $t = 0.54$, $TP = 5$ and $TN = 9$, so accuracy $= 14/20 = 0.70$.

(c) For each positive, count the negatives scored below it: 10, 10, 9, 9, 9, 7, 6, 5, 2, 1, summing to 68. There are $10 \times 10 = 100$ positive–negative pairs, so AUC $= 68/100 = 0.68$.

Check via the ROC staircase: at each negative, record the current TPR: 0.2, 0.5, 0.5, 0.6, 0.7, 0.8, 0.8, 0.8, 0.9, 1.0. These sum to 6.8; multiplying by the step width 0.1 gives 0.68.

(d) The perfect classifier is at the top-left corner $(0, 1)$ with AUC $= 1$. A random classifier lies on the diagonal TPR $=$ FPR with AUC $= 0.5$.